



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Statistical Diagrams

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Your notes

Stem & Leaf Diagrams

Stem & Leaf Diagrams

What is a stem-and-leaf diagram?

- A **stem-and-leaf diagram** is a simple way to display an **ordered list** of data using **digits**
- Two-digit numbers are split into a tens digit (the **stem**) and a units digit (the **leaf**)
 - 25 becomes 2 | 5
 - The stem is written **vertically** and the leaves are written **horizontally** (in order)
- The following diagram shows the ages below
 - 11, 18, 20, 21, 25, 28, 29, 35, 36, 40

	Age
1	1 8
2	0 1 5 8 9
3	5 6
4	0

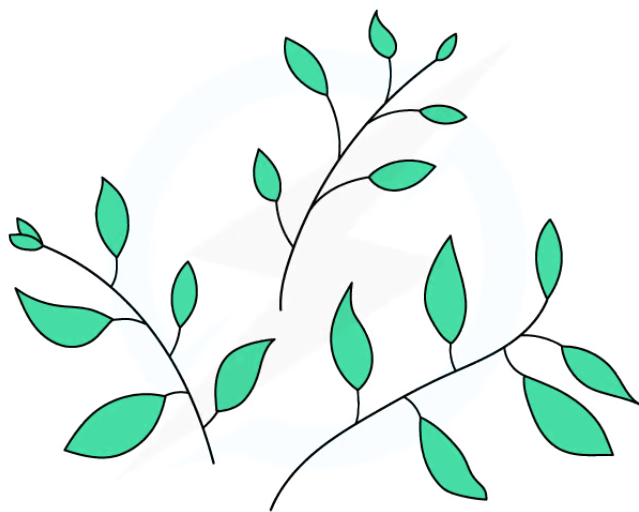
Key: 1|8 means 18 years old

What is the key on a stem-and-leaf diagram?

- The **key** shows how values are **formed** from digits
 - It should include **units**
- Other keys are possible
 - 2 | 5 represents 2.5 degrees
 - 2 | 5 represents 2005 people



Your notes



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How do I find the median from a stem-and-leaf diagram?

- The **median** is the **middle** number
 - Data values are already in **order**
 - You can cross out numbers from the beginning and end until they meet in the middle
 - Remember that the **highest** number will be at the **end** of the last stem
 - If two numbers remain in the middle, find the **midpoint** between them



Examiner Tips and Tricks

A common mistake when finding the median is to write down the leaf only (median = 6) rather than the stem and leaf (median = 26).



Worked Example



Your notes

A hospital is investigating a new drug that claims to reduce blood pressure.

The reductions in blood pressure, measured in mmHg (millimetres of mercury), for 11 patients are shown below.

12 31 24 18 21 34 40 19 23 17 16

(a) Draw a stem and leaf diagram to show these results.

Split each value into its tens digit (stem) and units digit (leaf)

The values are not yet in order

	Blood pressure reduction
1	2 8 9 7 6
2	4 1 3
3	1 4
4	0

Put the values in order

Write down the key

	Blood pressure reduction
1	2 6 7 8 9
2	1 3 4
3	1 4
4	0

Key: 1|2 means a blood pressure reduction of 12 mmHg

(b) Use your stem and leaf diagram to find the median blood pressure reduction.

Find the middle value

This is the 6th patient

	Blood pressure reduction
1	2 6 7 8 9
2	(1) 3 4
3	1 4

4	0
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The median has a leaf of 1 and a stem of 2

The median is 21 mmHg



Your notes

Bar Charts & Pictograms



Your notes

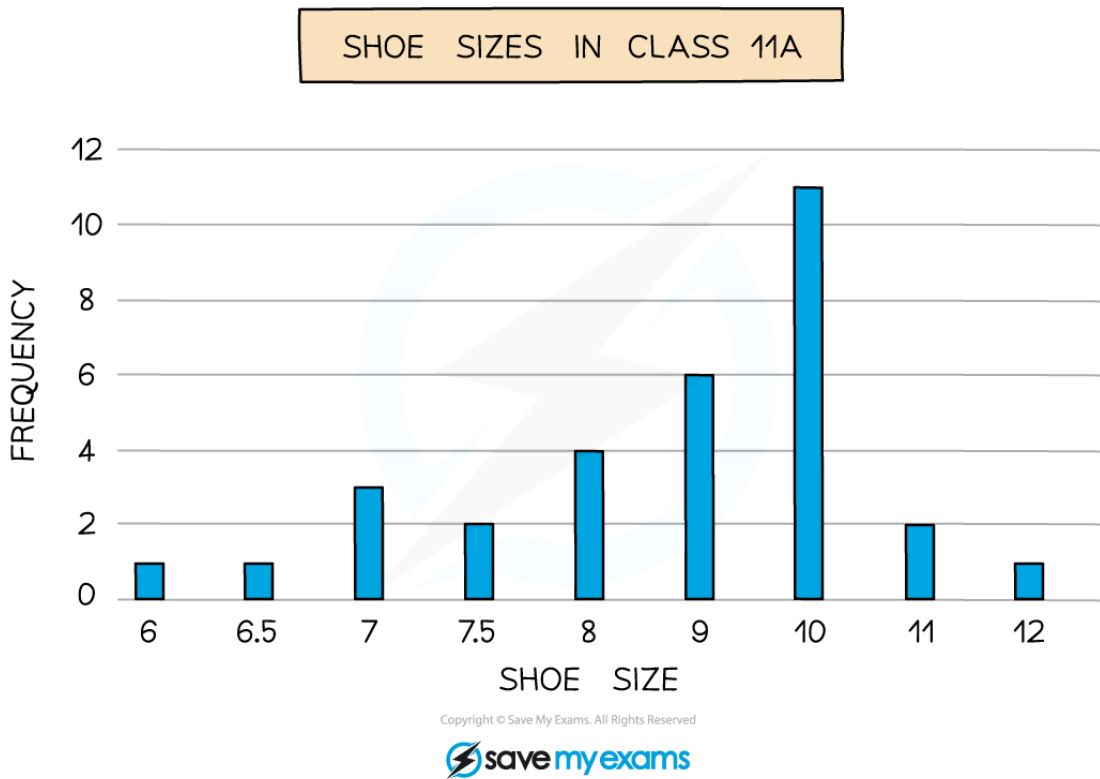
Bar Charts & Pictograms

What is a bar chart?

- A **bar chart** is a visual way to represent **discrete** data
 - Discrete data is data that can be **counted**
 - This can be **numerical** like shoe sizes in a class
 - Or **non-numerical** (categorical) like colours of cars down a road
- The **horizontal axis** shows the different **outcomes**
- The **vertical axis** shows the **frequency**
- The **heights** of the **bars** show the frequency
 - Bars should be **separated** by gaps
 - Bars should have **equal widths**



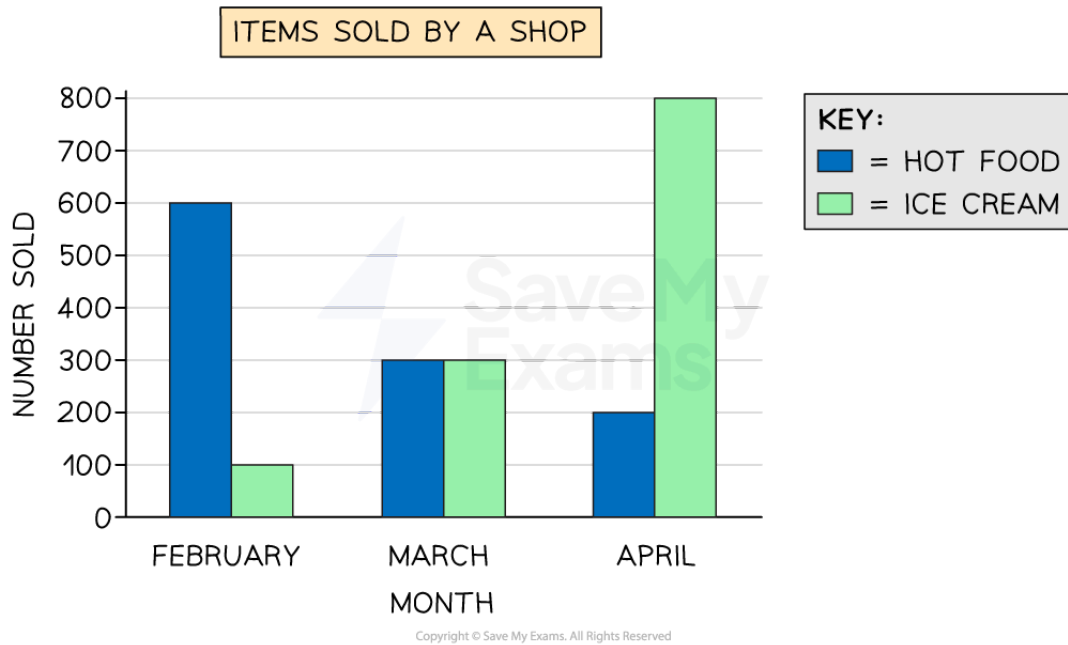
Your notes



- The **mode** is the **outcome** with the **highest** bar
- You can also get **dual bar charts** to compare two data sets
 - Bars are in **pairs** (side-by-side) for each outcome



Your notes

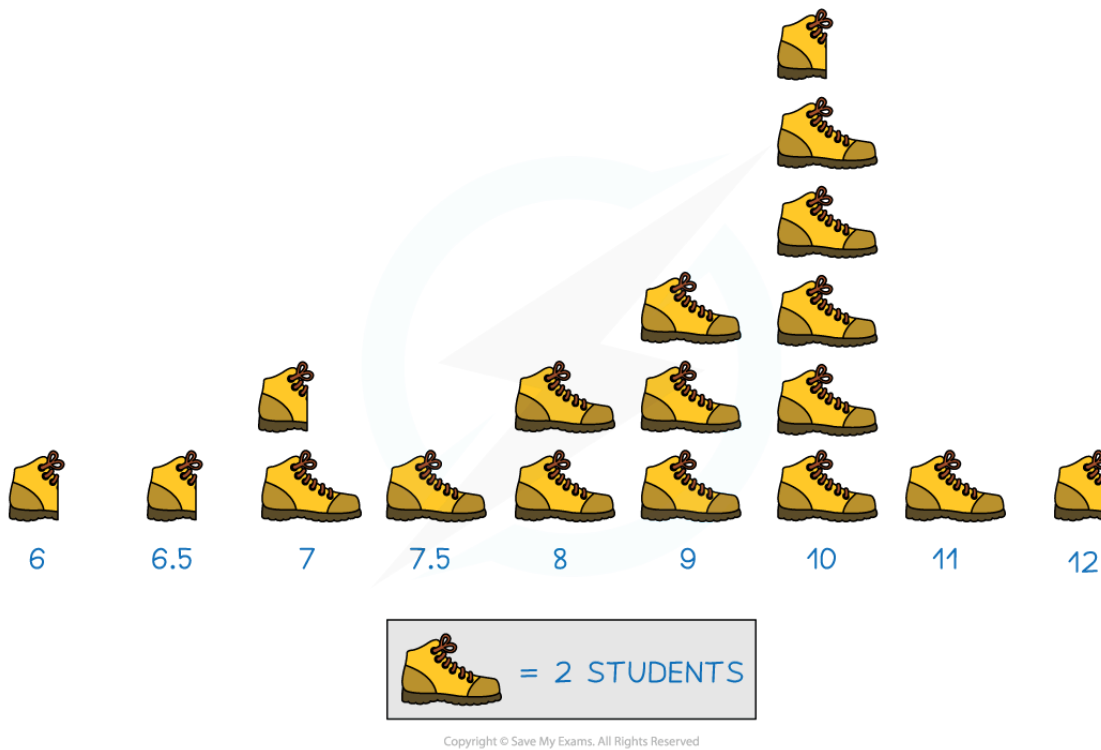


What is a pictogram?

- A **pictogram** is an **alternative** to a bar chart
 - It is used in the same situations
- There are **no axes**
 - **Frequency** is represented by **symbols**
 - A **key** shows the value of 1 symbol
 - For example, 1 symbol represents a frequency of 2
 - Half and quarter symbols are often used



Your notes



- The pictogram above shows the shoe sizes of students in a class
 - As 1 picture of a shoe represents 2 students
 - Half a shoe represents 1 student
 - The number of students with a shoe size of 7, is 3



Examiner Tips and Tricks

- If asked to draw a bar chart, find the largest frequency and choose a scale which makes that fit in the space provided
- If asked to draw a pictogram, pick a symbol that is easy to duplicate and draw half (or quarter) of

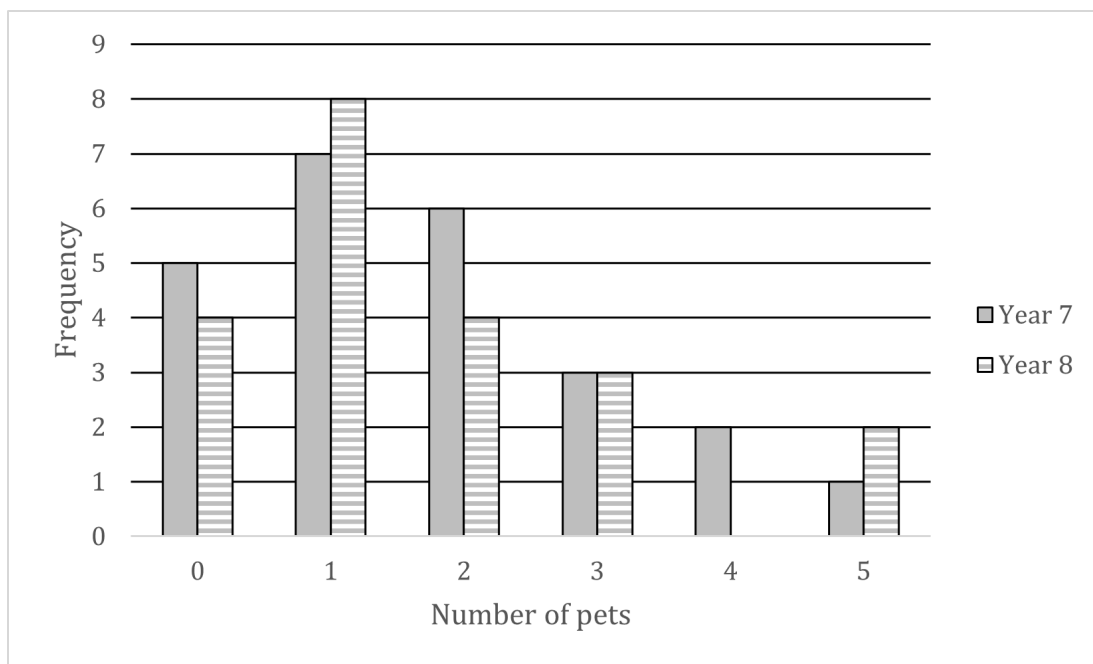


Worked Example



Your notes

Mr Barr teaches students in Year 7 and Year 8.
He records the number of pets that students in each year have.
His results are shown below.



(a) Write down the modal number of pets for his Year 7 students.

The modal number (mode) is the number of pets that occurs the most
Visually, this will be the highest bar for Year 7s

The mode for Year 7 is 1 pet

(b) How many Year 8 students does he teach?

Add up all the heights (frequencies) of the Year 8 bars

$$4 + 8 + 4 + 3 + 0 + 2$$

He teaches 21 Year 8 students



Your notes

Pie Charts

Pie Charts

What is a pie chart?

- A **pie chart** is a circle divided into sectors which is used to **present data**
- The **sectors** represent different **categories**
 - They show the relative **proportions** of the categories
 - They do **not** show the actual **frequencies** of each category

How do I draw a pie chart?

- This is easiest explained through an example
- The following data shows the favourite colours of a class of students

Colour	Red	Purple	Blue	Green	Yellow	Orange
Students	11	4	9	3	2	1

- Write the frequencies as **fractions** of the **total number** of students, 30

Colour	Red	Purple	Blue	Green	Yellow	Orange
Students	11	4	9	3	2	1
Fractions	$\frac{11}{30}$	$\frac{4}{30}$	$\frac{9}{30}$	$\frac{3}{30}$	$\frac{2}{30}$	$\frac{1}{30}$

- Find the **angles** of the sectors by **multiplying each fraction** by **360°**
 - Then check all angles add up to 360°

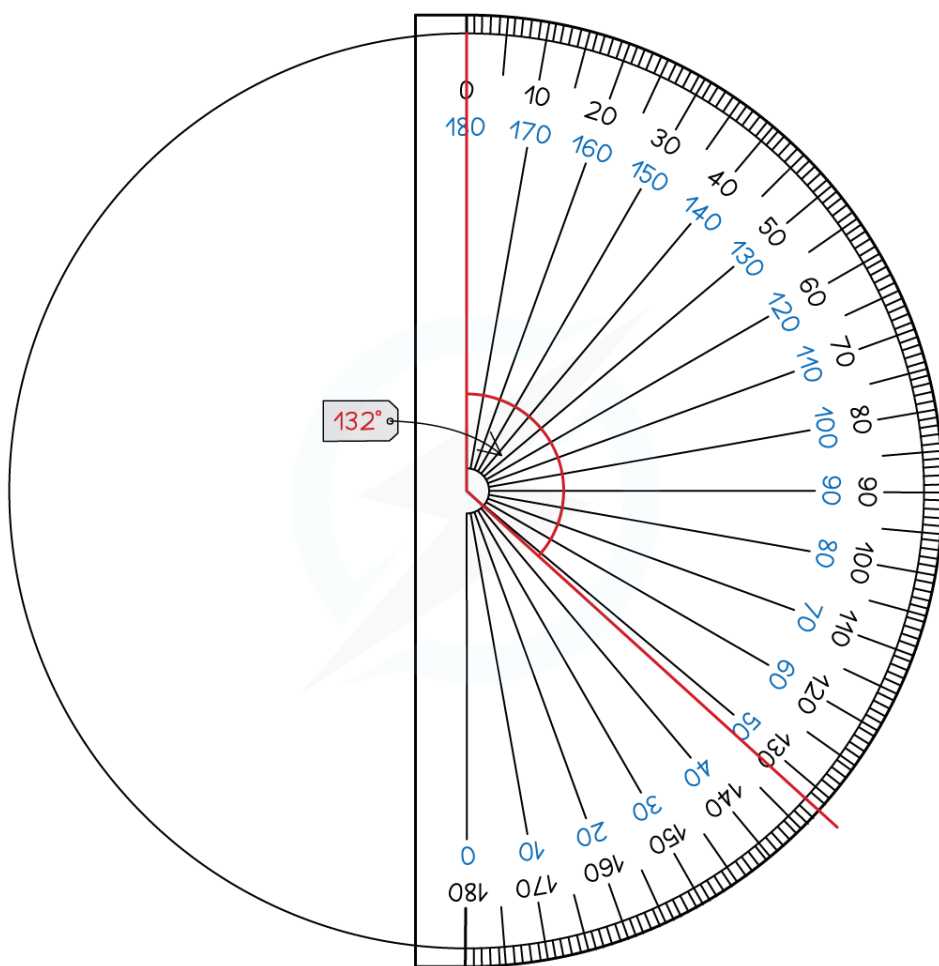
Colour	Red	Purple	Blue	Green	Yellow	Orange
Students	11	4	9	3	2	1



Your notes

Fractions	$\frac{11}{30}$	$\frac{4}{30}$	$\frac{9}{30}$	$\frac{3}{30}$	$\frac{2}{30}$	$\frac{1}{30}$
Angles	132°	48°	108°	36°	24°	12°

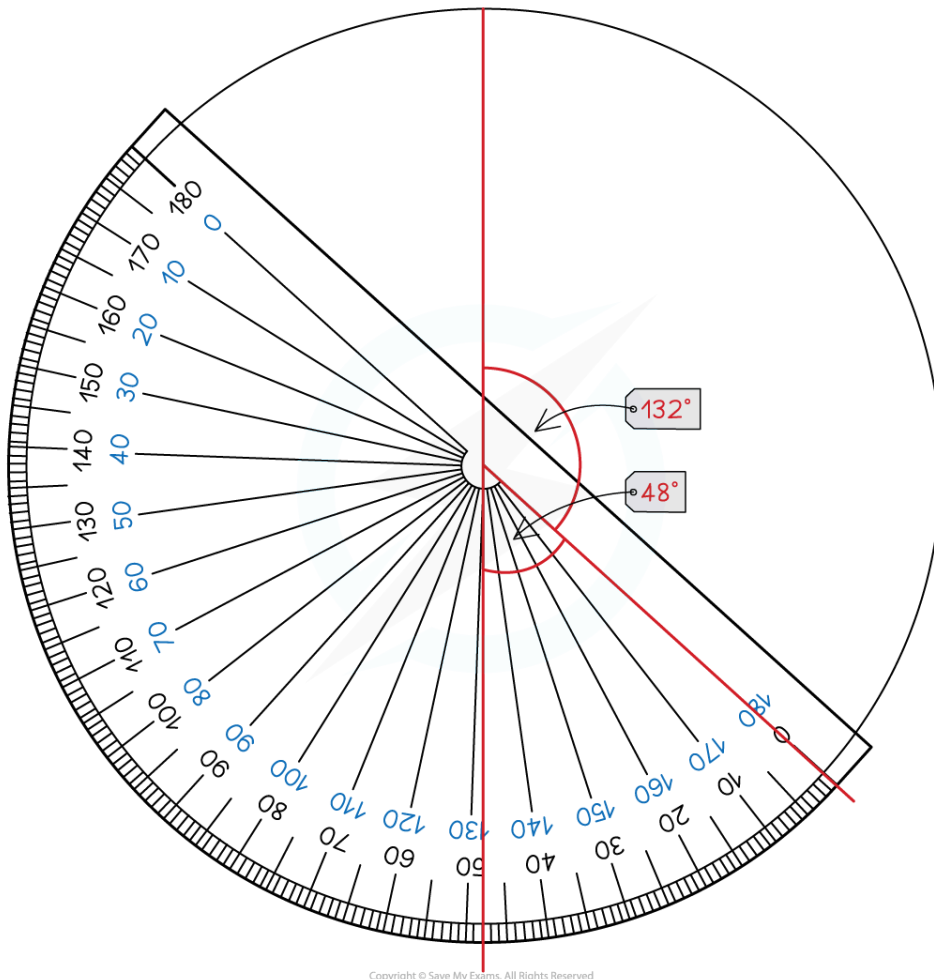
- Draw a vertical line from the circle's centre to the top, then use a **protractor** to mark off the first angle
 - Draw a line from the centre to this first mark
 - Then, from this line, mark off the next angle (and so on)



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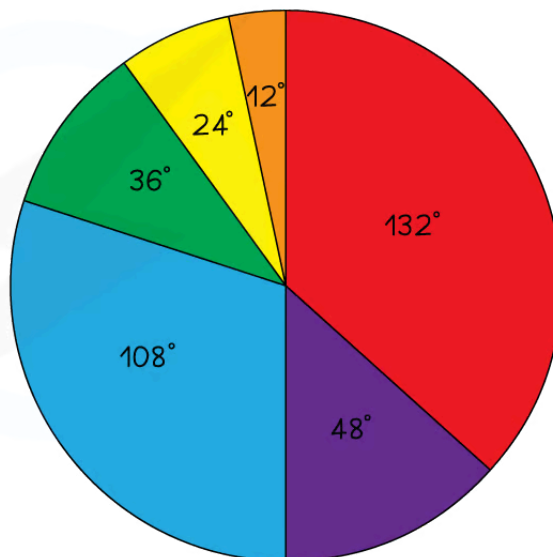
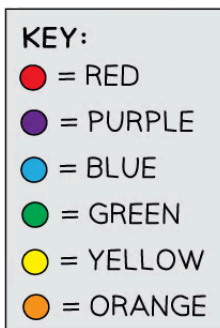
Your notes





Your notes

STUDENTS' FAVOURITE COLOUR



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How do I solve problems with pie charts?

- Use the following facts
 - **angles** are **proportional** to the **frequencies** of each category
 - **360°** represents the **total frequency**
- For harder problems, it helps to work out
 - **what frequency** is represented by **1°**
 - **what angle** is represented by **1 unit** of **frequency**
- For example, if a sector of 30° represents 15 people, then
 - 1° = 0.5 people (dividing by 30)
 - 2° = 1 person (multiplying by 2)
- These relationships can then be scaled **up** or **down** accordingly
 - If 1° = 0.5 people

- then $360^\circ = 180$ people (multiplying by 360)



Examiner Tips and Tricks

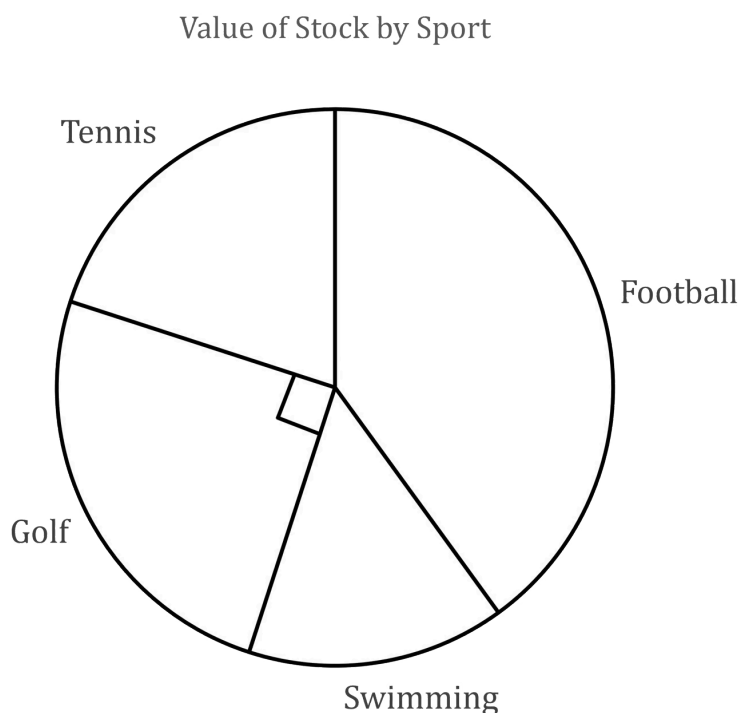
- If the pie chart says 'not to scale', then examiners want you to use ratio and proportion methods to answer the questions
- Don't measure angles using a protractor!



Worked Example

The following pie chart represents the values of items stocked in a sports shop.

[not to scale]



(a) Given that the shop stocks \$12 000 of golf items, find the total value of the shop's stock.

Find a relationship between an angle and a value



Your notes



Your notes

$$90^\circ = \$12\,000$$

The total value is represented by 360°

Multiply by 4 to get from 90° to 360°

$$360^\circ = 3 \times 12\,000$$

Total value is \$48 000

You can also do this question by finding 1° first

(b) If the angle on the pie chart for tennis is 72° , find the value of tennis items that are stocked by the shop.

It is quickest to find the fraction $\frac{72}{360}$ of the total value, found in part (a)

$$\frac{72}{360} \times 48\,000 = 9\,600$$

The value of tennis items is \$9 600



Your notes

Reading & Interpreting Statistical Diagrams

Reading & Interpreting Statistical Diagrams

How do I interpret statistical diagrams?

- Read and **understand** the initial sentences describing the **situation** (context)
 - **Underline** important words if necessary
- Look for any **keys** that may help you to understand the diagram
 - For example
 - 1 unit represents 20 people
 - Year 10 is the **solid** line, Year 11 is the **dotted** line
 - Class A is **shaded**, class B is **striped**
- **Read** the **titles** of diagrams and all **axes labels**
 - A graph for '**new** students at a school' is different to '**all** students at the school'
 - A **frequency** axis that **starts at 50** is different to one starting at 0
- Understand which **units** are being used
 - Individual lengths may be in **centimetres** but total length may be in **metres**
 - Populations may be measured in **thousands**
- Look out for any **extreme values** (outliers / anomalies)
 - One month's temperature might be unusually high
 - Was it a heat wave or a recording error?

How do I draw conclusions from diagrams?

- Look for **overall trends** in the diagram
 - Prices **increase** year on year
 - The temperature **peaks** in June
- Use **numbers** from the graphs
- Refer to any **changes**

- The **steepness** (gradient) of graph may change
- Write in **full sentences** that copy the **exact wording** from the question
 - 'The number of goats in farm A has decreased by 12 over the 8 month period'
 - **Not** 'There are fewer of them now'
- You may need to calculate the **mode, median, mean** or **range** to support any explanations
- Understand why drawing conclusions may **not** be **suitable**
 - The data set may be **too small** to be representative
 - The data set may be **biased**
 - Consider the **scope** of the data
 - e.g. Data for January to March cannot be used to predict August



Your notes

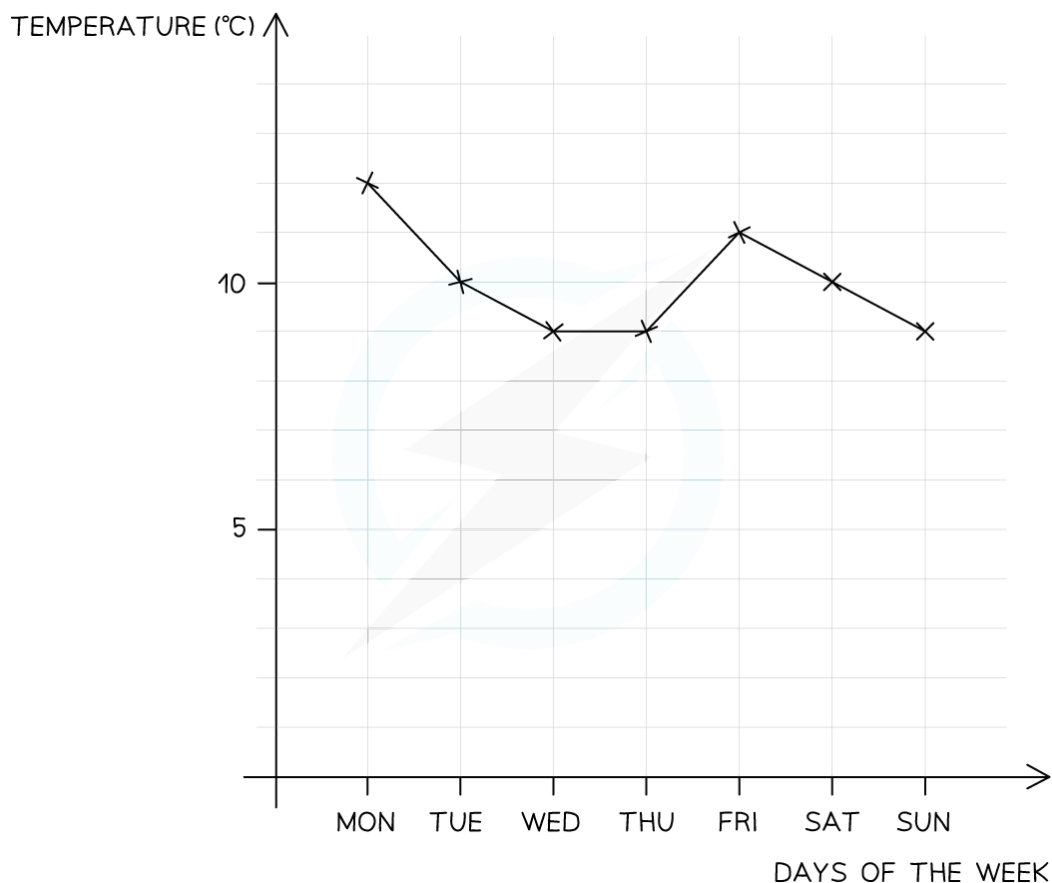


Worked Example

The diagram below shows the temperature of a garden in the UK, recorded at 7am on each day of a particular week in March.



Your notes



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(a) Describe the change in temperature over the first four days.

The trend shows a decrease in the first three days, then a constant temperature

Find numbers from the graph to use in your answer

Refer to the steepness of the change

The temperature decreases from 12°C on Sunday to 9°C on Wednesday

The decrease is steeper over the first two days

There is then a constant temperature of 9°C on both Wednesday and Thursday

(b) A gardener claims that, based on the graph, Monday must have experienced the highest temperature that week.

Give a reason as to why this might not be true.

Reread the information at the start

These temperatures were recorded at 7am in the morning (we don't know how hot the rest of the day was)



Your notes

The temperatures on the graph are at 7am each day
The maximum temperature may have been after 7am, on a day that was not Monday

(c) A journalist wants to use the data shown to claim that the average temperature that week was below 10°C .

The mean of the temperatures shown is 10°C .

Which type of average would you suggest they use? Explain your answer.

The mean of 10°C does not support the claim that the average temperature is below 10°C

Try calculating the mode instead

12, 10, 9, 9, 11, 10, 9

The most frequent number is 9

The modal temperature is 9°C
 $9^{\circ}\text{C} < 10^{\circ}\text{C}$ so using the mode would help the journalist's claim

You could try the median, but it is also 10°C



Your notes

Comparing Statistical Diagrams

Comparing Statistical Diagrams

How do I compare statistical diagrams?

- You may be given **two graphs** for two different **data sets** with the same **context**
- Compare **trends** in the graphs
 - **Increases, decreases, maximum points**
 - **Steepness** of the change
- Comment on **differences** and **similarities**
- Explain **clearly** which **part** of the graph you are talking about
- Use **numbers** from each data set in your comparisons
- Comment on any **extreme values** (outliers)
- Calculate **means, medians** or **modes** to compare **averages**
- Calculate **ranges** to compare the **spread** of values



Examiner Tips and Tricks

- Always relate numbers and calculations back to the context of the question
 - If you don't, you won't score all the marks!

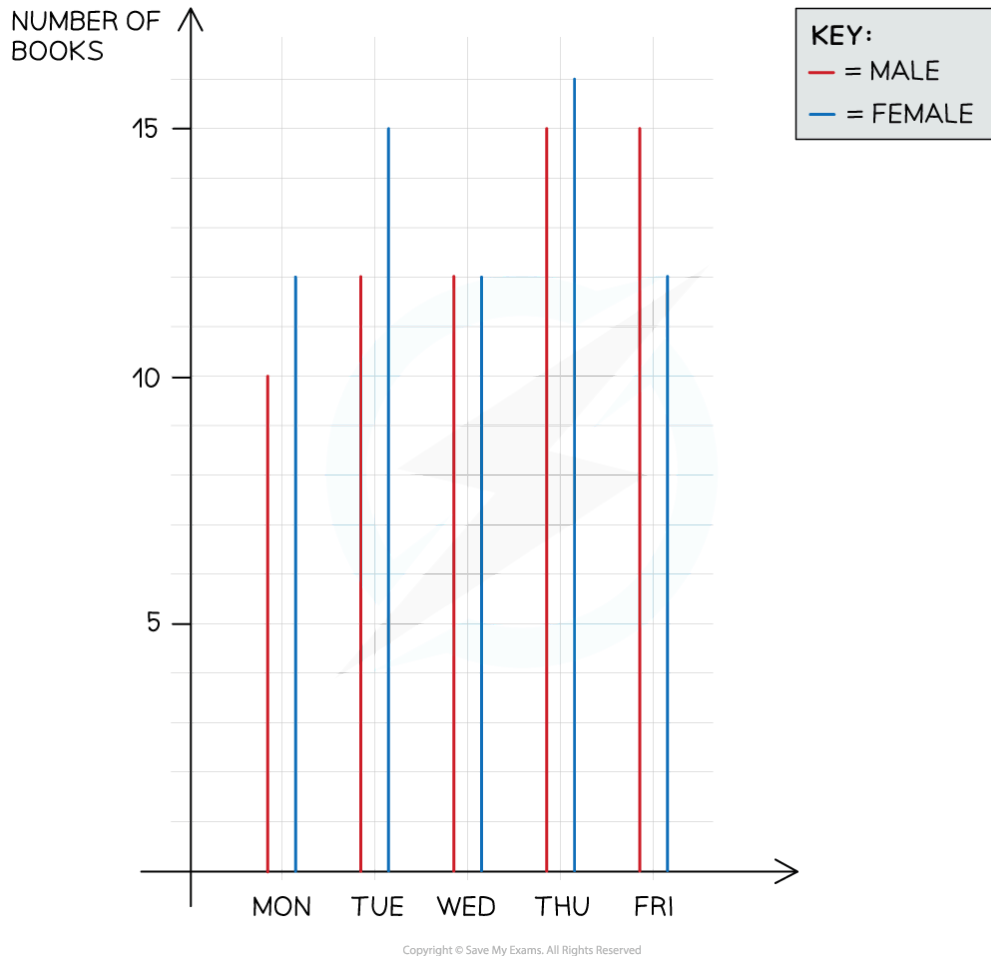


Worked Example

The number of books bought during the opening week of a new bookshop is shown below.
The shopkeeper wants to investigate shopping patterns between male and female customers.



Your notes



(a) Briefly summarise the overall trend in the number of books bought by male customers.

Look at the heights of the red lines

Comment on how they increase ('briefly' means you do not have to talk about every point)

Use numbers from the graph and exact phrases from the question

The number of books bought by male customers increases from 10 on Monday up to 15 on both Thursday and Friday

(b) The shopkeeper says that the changes in the blue lines suggest that the amount of books bought by females varies more than males.

By comparing a suitable statistical measure, show that this is not necessarily true.



Your notes

Variations refers to the spread of data, so compare their ranges
For male and female, subtract the smallest value from the biggest value

Male range is $15 - 10 = 5$

Female range is $16 - 12 = 4$

Compare the numbers
Explain how this relates to the spread of the data
State how your findings do not agree with the shopkeeper

The range of male customers is greater than that of female customers, $5 > 4$
This suggests male customers have a greater spread (variability) in books bought
This is the opposite of what the shopkeeper said

(c) Give one reason as to why the shopkeeper should not use the data shown to draw conclusions about shopping patterns.

Reread the sentences at the beginning
This data is for the opening week of the bookshop only
State that this is unrepresentative of a normal week
Give a specific real life example

The data shown is for the opening week of the bookshop
It is unlikely to be representative of a normal week
The numbers may increase as the bookshop becomes more popular

Or decrease if the customers lose interest
(Any reasonable example would work)